PHOTO ESSAY

Multilobular Choroidal Melanoma Following Ruthenium 106 Brachytherapy

Choroidal melanoma usually presents as a monocular, monofocal tumor that is dome shaped or mushroom shaped. Occasionally, choroidal melanoma is bilobal and rarely, it is multilobular.[1] The present case is a unique multilobulated choroidal melanoma.

An 82-year-old woman sought care at our clinic because of a suspicious macular lesion in her right eye that caused deterioration of the visual acuity to light perception. The visual acuity in the left eye was 0.4 (6/15). An elevated mass, melanotic in its base and amelanotic in its apex, was seen in the macula of the right eye, with some retinal detachment surrounding it. The fundus of the left eye was normal. Ultrasound examination showed a dome-shaped solid mass in the temporal side of the macula, measuring 10.3 mm × 10.4 mm in diameter and 6.1 mm in maximal height, with low internal reflectivity, rimmed by detached retina. The tumor was treated by ruthenium 106 brachytherapy, irradiating 10 200 cGy to the apex and 70 000 cGy to the base of the tumor.

In follow-up examination at 1, 2, and 4 months after the treatment, no significant change was noticed. On examination 9 months after the brachytherapy, a marked increase in the tumor size was noticed. A multilobulated tumor was seen, composed of a pigmented base and many amelanotic lobules over it, pushing the retina aside (**Figure 1**). In ultrasound examination, a multilobulated lesion was seen with maximal height of 10.3 mm (**Figure 2**). The eye was enucleated soon after the examination and was sent for histopathologic evaluation.

Gross examination of the eye, after opening it horizontally, revealed a fungating tumor in the posterior pole. Its maximal diameter was about 10 mm and its maximal height 10 mm. The retina over the tumor was detached. Microscopic examination revealed a lightly pigmented multilobulated tumor in the temporal part of the posterior pole, breaking through the Bruch membrane. The tumor was composed of epithelioid cells (**Figure 3**), and 5 mitotic figures were counted in 20 high-power fields. The proliferation marker MIB-1 was used to stain for Ki-67, which showed marked proliferative activity. No necrotic areas were identified in the tumor.

COMMENT

The multilobular uveal melanoma of our patient represents a highly unusual response to radiation by ruthenium 106 brachytherapy. It is well known that after radiation, the tumor's remnant shows viable cells and even cell proliferation.[2] In our experience, posterior uveal melanomas that were enucleated because of regrowth after brachytherapy had a similar proliferation activity like tumors that were enucleated without being irradiated.[2] We can hypothesize that in our case, incomplete therapy left some tumor cells partially treated, which allowed radiation-induced mutation in the surviving tumor cells that led to outgrowth of more aggressive parts. In some cases, regrowth of uveal melanoma was independently related to risk of tumor-related death.[3]

The multilobulation of the tumor in our patient is difficult to explain. There were no multiple breaks in the Bruch membrane around any of the lobules, but only around the neck close to the tumor base. No necrosis, which could explain the unique shape of the tumor, was identified in any part of the tumor. The only explanation that we can hypothesize is a different speed and direction of growth in various parts of the tumor that were effected differently by the irradiation.

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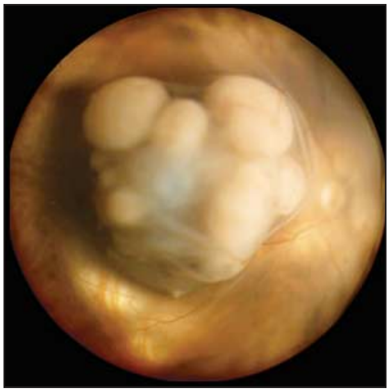

Figure 1. A multilobulated tumor with multiple amelanotic nodules over a pigmented base occupies the macular area of the right eye. The retina is pushed aside by the tumor.

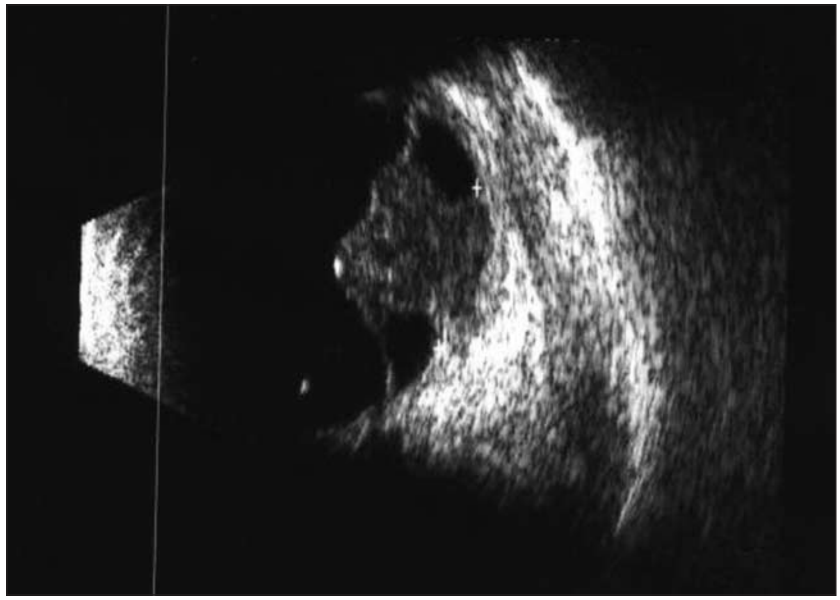

Figure 2. A B-scan ultrasound picture shows lobulated solid tumor surrounded by retinal detachment.

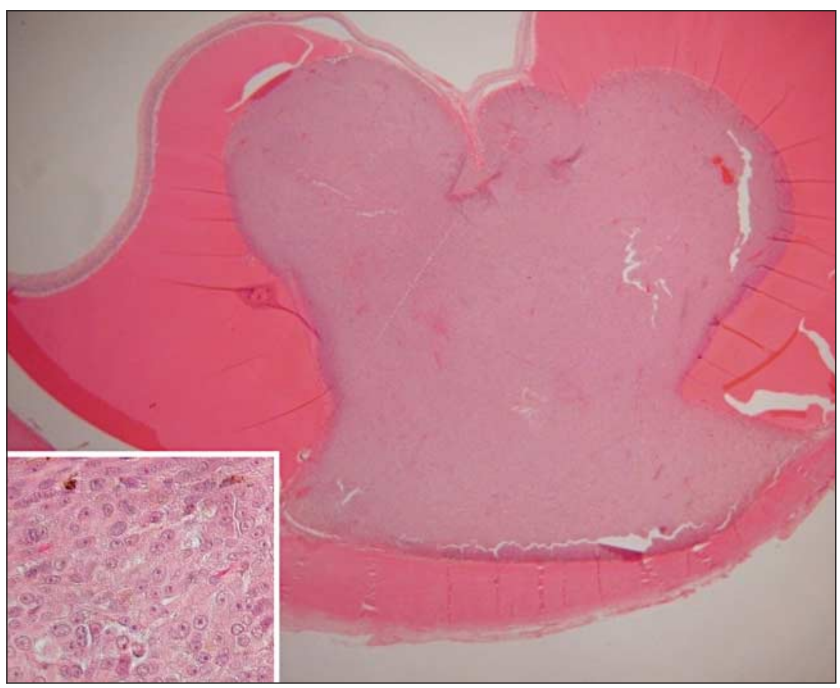

Figure 3. A whole-mount histological picture of multilobulated uveal melanoma surrounded by subretinal fluid. Histological examination reveals epithelioid-type melanoma (hematoxylin-eosin, original magnification × 100).